

# AI Literacy Glossary

## Week 1 - AI Fundamentals

**Purpose.** This glossary captures the foundational vocabulary and ideas introduced in Week 1. Later weeks will build on these concepts without redefining them, so the glossary remains cumulative rather than repetitive.

**Week 1 focus:** what AI is, how major approaches developed, how models sit inside wider systems, how capability is measured, and how to interpret claims about current and future AI.

## Foundational concepts

### Artificial intelligence (AI)

Artificial intelligence is the broad umbrella term for computational systems designed to perform tasks involving capabilities associated with intelligent behaviour. These capabilities can include recognising objects, predicting outcomes, generating content, reasoning, planning, recommending choices and taking actions.

AI is not a single technology or product. The boundary around what counts as AI has also shifted historically: capabilities that once seemed remarkable can become ordinary software once they are familiar and reliable.

**AI is a field and a family of capabilities - not a synonym for ChatGPT or any other single product.**

### AI capability

An AI capability is something an AI system is able to do. Thinking in terms of capabilities is often more useful than beginning with product names or algorithms.

- Recognise - identify what something is.
- Predict - estimate what is likely to happen or what category something belongs to.
- Recommend - rank, prioritise or suggest alternatives.
- Generate - create new text, images, audio, video or code.
- Search and synthesise - locate, combine and summarise information.
- Act - carry out tasks in software or the physical world.

**A single AI application may combine several capabilities at once.**

### Conventional software

Conventional software generally obtains its behaviour from rules and logic explicitly specified by programmers. A simple conceptual picture is: input -> human-written rules -> output.

This contrasts with machine learning, where some useful behaviour is learned from examples rather than specified rule by rule. The distinction is conceptual rather than absolute because many real systems combine explicit rules with learned components.

#### Example

A tax calculator may contain explicit rules specifying how different income bands are treated.

### Machine learning (ML)

Machine learning is a major approach within AI in which useful behaviour is learned from data. Instead of specifying every rule required to turn an input into an output, developers provide data, examples, goals and a learning process from which a model is produced.

Week 1 uses machine learning mainly to establish this contrast with conventional software. The mechanics of how learning occurs are developed in Week 2.

## AI model

An AI model is the learned computational component that transforms inputs into outputs. A model might classify an image, estimate a risk, generate text, recognise speech or recommend an item.

The model is not the whole AI product. The same underlying model can sit inside different applications and systems with very different capabilities, safeguards and consequences.

**Model != application != wider AI-enabled system.**

## AI application

An AI application is the software interface or product through which people interact with one or more AI models. It may add conversation history, uploaded files, web access, safety rules, search, user-interface features or connections to other software.

Two applications can therefore use similar underlying models yet behave differently because the surrounding software provides different information and capabilities.

## AI-enabled system

An AI-enabled system is the wider arrangement surrounding an AI model. It can include the model, application, data, retrieval mechanisms, tools, permissions, people and organisational processes.

This wider-system view matters because real-world consequences often depend as much on what the system can access and do as on the raw capability of the model itself.

### Example

A language model that only suggests wording in a draft has very different consequences from a similar model connected to customer databases, payment systems and permission to take actions.

## Capability, access and authority

These are three different dimensions of an AI-enabled system. Capability asks what the model can infer, generate, predict or plan. Access asks what information, software and tools the system can reach. Authority asks what actions the system is permitted to take.

A relatively modest model with extensive access and authority can create greater real-world consequences than a more capable model isolated inside a low-stakes chatbot.

**Capability x access x authority = consequence.**

## Human oversight

Human oversight refers to the ways people supervise, constrain, review or intervene in an AI system's behaviour. Oversight becomes increasingly important as systems gain tools, access and permission to act.

Oversight can be built into a system through approval requirements, restricted permissions, monitoring, escalation paths and limits on what actions can be performed. Later weeks develop this idea in more detail.

# Contemporary AI families and system properties

## Deep learning

Deep learning is a major branch of machine learning based on neural networks that can learn complex internal representations from data. Earlier machine-learning systems often depended heavily on humans deciding

which features should be presented to the learner; deep learning increasingly allows useful representations to be learned directly from images, text, audio and other data.

Its rapid progress in the 2010s was enabled by a combination of improved algorithms, much larger datasets and greatly increased computing power.

## Representation

A representation is an internal way of encoding information so that it can be used by a computational system. A useful representation preserves aspects of the input that matter for a task while ignoring much irrelevant detail.

The ability to learn useful representations is one of the major reasons deep learning became powerful across perception, language and other complex domains.

### Example

A visual system does not have to treat an image as millions of unrelated pixels; it can learn internal representations that capture shapes, textures, objects and higher-level visual structure.

## Generative AI

Generative AI refers to AI systems designed to produce new content such as text, images, audio, video or code. It differs from a purely discriminative system that only assigns an existing category or score.

Generative AI is largely enabled by modern deep-learning and foundation-model approaches. Week 3 examines in detail how generative systems produce content and why that content can be variable or inaccurate.

## Foundation model

A foundation model is a broadly trained model whose learned capabilities can be reused across many downstream tasks rather than being built for only one narrow purpose.

This shift from one specialised model per task towards reusable capability is one of the defining developments of contemporary AI.

### Example

The same broadly trained model may support writing, translation, analysis, coding, summarisation and perception instead of requiring a separate model for every task.

## Transfer

Transfer is the reuse of capabilities or representations learned in one context for other tasks or domains.

Foundation models are important partly because broadly learned representations can transfer to tasks they were not individually trained from scratch to perform.

Transfer does not imply perfect performance everywhere; it means that useful structure learned in one setting can support performance elsewhere.

## Multimodal AI

Multimodal AI refers to systems that can work across more than one type of information, or modality.

Common modalities include text, images, audio, video and sensor data.

Multimodality matters because real-world problems rarely arrive neatly as only one kind of input.

### Example

A multimodal system might inspect a photograph, read accompanying text and answer questions that require information from both.

## Agentic AI / agentic system

An agentic AI system is designed not merely to generate an answer but to pursue goals through sequences of actions. It may combine models with memory, tools, planning, feedback and repeated action.

Agency is mainly a property of the wider system rather than a distinct model type. A model becomes part of an agentic system when it is wrapped in mechanisms that allow it to maintain state, use tools and act over time.

## Historical paradigms and changing bottlenecks

### Symbolic AI

Symbolic AI represents knowledge, concepts and relationships explicitly and manipulates them using rules, logic and search. It can work very effectively when the relevant concepts and rules can be represented cleanly.

Its difficulty is that the real world contains ambiguity, uncertainty, exceptions and interacting variables that can make large rule systems difficult to construct and maintain.

#### Example

IF road is wet AND temperature is below freezing, THEN warn about ice.

### Expert system

An expert system is a knowledge-based AI system that attempts to reproduce specialised decision-making by encoding expert knowledge into a structured collection of rules.

Expert systems achieved meaningful success in narrow domains, but large systems were difficult to scale because acquiring expert knowledge, maintaining rule bases and handling unexpected cases became major bottlenecks.

### Statistical AI / statistical learning

Statistical learning shifted AI away from requiring people to explicitly describe every rule and towards learning probabilistic relationships from observations. Probability, prediction and uncertainty became central.

Conceptually, the question changed from “What rules describe the world?” to “What patterns can be learned from observations?”

#### Example

Spam detection, credit-risk estimation, demand forecasting and image classification are examples of problems that can be approached statistically.

### Computational bottleneck

A bottleneck is the factor currently limiting further progress on a problem. The history of AI can partly be understood as a succession of changing bottlenecks.

For symbolic AI, representing messy real-world knowledge became a bottleneck. For earlier machine learning, manually designing useful representations was often a bottleneck. Deep learning reduced some of these constraints but created or exposed others involving data, compute, reliability and scale.

**A useful historical question is: which bottleneck has moved, and what new bottleneck has replaced it?**

### Compute

Compute refers to the computational resources required to train and operate AI systems. In practice this includes specialised processors, servers, data centres, electricity and supporting infrastructure.

Compute became especially important to modern deep learning because many useful ideas existed before sufficient computational capacity was available to apply them at large scale.

## Measuring AI capability

### Benchmark

An AI benchmark is a standardised collection of tasks used to measure and compare system performance. A useful analogy is an exam for an AI system.

Benchmarks help researchers compare models, track progress and identify weaknesses. Like an exam, however, a benchmark samples only part of competence. High performance on one benchmark does not establish general intelligence or reliability everywhere.

### Benchmark score

A benchmark score is the numerical result produced when a system is evaluated against a benchmark. The score is evidence about performance on the tasks that were tested, not a complete measurement of intelligence.

Interpreting a score requires asking what was tested, how broad the test was, whether the benchmark still distinguishes between systems, and how closely it resembles the real-world use of interest.

### Benchmark saturation

Benchmark saturation occurs when high-performing systems cluster close to the maximum score, so the benchmark becomes less useful for distinguishing between them.

Saturation can be a sign of genuine progress, but it can also be influenced by repeated optimisation against a familiar benchmark, exposure to similar material, or a benchmark that is too small or coarse for current systems.

#### Example

If almost every student scores between 98% and 100% on an exam, the exam may still show mastery but has become poor at distinguishing who is stronger.

### Benchmark contamination

Benchmark contamination occurs when benchmark questions, answers or closely related material become represented in data used to train or develop a model.

This can make evaluation results look stronger than the model's ability to solve genuinely unfamiliar problems. Contamination is one possible explanation for inflated performance, not an automatic explanation for every strong score.

### Jagged intelligence

Jagged intelligence describes the highly uneven pattern of capabilities exhibited by contemporary AI systems. A system can perform exceptionally well on a sophisticated task yet fail on another task that humans consider much simpler.

The key mistake is assuming that AI capability is distributed across tasks in the same way human capability is. Exceptional performance in one area should not automatically be generalised into a claim of broad competence.

# Claims about intelligence

## Performance

Performance refers to what a system demonstrably succeeds or fails at doing on a task. It is an empirical question: for example, whether the system answered correctly, completed a task or achieved a particular score.

Strong task performance should not automatically be equated with deeper claims about understanding, agency, consciousness or general intelligence.

## Understanding

Understanding is a stronger and more contested claim than successful task performance. It concerns what kinds of internal representations or processes support a system's behaviour and whether those processes deserve to be described as understanding.

Week 1 does not settle this philosophical and scientific debate. The important distinction is that “the system performed the task successfully” and “the system understands the task” are different claims.

## Agency

Agency refers to the capacity of a system to pursue goals and take actions over time. It differs from simply producing a response.

A system becomes more agentic when it can maintain a goal, plan, use tools, monitor progress, react to feedback and take repeated actions. Agency is therefore closely tied to the wider system around a model.

## Consciousness

Consciousness generally refers to subjective experience - whether there is something it is like to be the system. Whether any current AI system is conscious remains scientifically and philosophically disputed.

Fluent language, difficult problem solving or apparently intelligent behaviour do not by themselves establish consciousness.

## Artificial General Intelligence (AGI)

Artificial General Intelligence, or AGI, is a contested term generally used for AI with broad capability across many domains rather than competence in only a narrow set of tasks.

There is no universally accepted technical threshold or test for AGI. Discussions may refer to breadth across domains, human-comparable performance, transfer between tasks, reasoning or autonomy, and these properties should not automatically be treated as equivalent.

**Broad capability, human comparison, agency, understanding and consciousness are separate questions.**

# Interpreting claims and forecasts

## Observation

An observation is an empirically grounded statement about what has happened or what can currently be measured. Examples include a benchmark score increasing, tool-use performance improving, systems becoming multimodal or costs changing.

Observations can still involve measurement uncertainty, but they describe evidence about the present or past rather than a claim about what must happen next.

## Extrapolation

Extrapolation extends observed trends beyond the available evidence to make claims about what may happen next. It always adds assumptions.

For example, “model performance increased substantially over the last three years” is an observation; “therefore the same trajectory will continue until machines exceed humans at almost everything” is an extrapolation.

**Observation tells us what the evidence shows. Extrapolation tells us what someone thinks the trend implies.**

## Assumption

An assumption is something accepted as part of an argument without being established directly by the available evidence. Forecasts about AI often depend on assumptions about future algorithms, compute, data, energy, regulation, economics, adoption and technical bottlenecks.

Making assumptions explicit helps turn an apparently definitive claim into something that can be examined and challenged.

## Conditional forecast

A conditional forecast expresses a prediction together with the assumptions on which it depends. This is usually more informative than presenting a date or outcome as certain.

The purpose is not to make forecasts verbose; it is to expose what would have to remain true for the forecast to be credible.

### Example

Instead of “AGI will arrive in 2030”, a conditional forecast would state that such a prediction depends on a particular definition of AGI, continued architectural progress, sufficient compute and data, and the absence of a dominant new bottleneck.

## AI forecast

An AI forecast is an estimate about future AI capability, adoption or impact. Forecasting is difficult because AI development depends on many interacting systems at once, including algorithms, computing resources, data, energy, capital, regulation, human adoption and scientific discoveries.

A prediction about future AI capability therefore contains implicit predictions about several other technological, economic and social systems.

# AI as an economic and institutional system

## Commercial incentive

A commercial incentive is an economic reason for an organisation to favour particular decisions, investments or ways of presenting information. Frontier AI developers may simultaneously be researchers, product vendors, infrastructure buyers, fundraisers and competitors for customers and talent.

Commercial incentives do not make a claim false. They do change the context in which evidence is selected, framed, amplified and repeated.

**The literacy question is not “Can I trust companies?” but “What evidence supports this claim, who produced it, and what interests surround it?”**

## Capability narrative

A capability narrative is the broader story used to interpret evidence about what AI systems can do and what that evidence means for the future.

The same empirical improvement can support different narratives: that current systems are rapidly approaching more general capability, that they remain structurally limited, or that new architectures will be needed. The key is to separate the observed evidence from the interpretation attached to it.

## Frontier AI

Frontier AI refers informally to the most capable or technically advanced AI systems being developed at a particular point in time. The frontier is not a fixed technology: it moves as new capabilities emerge and new bottlenecks become important.

Current frontier directions include more reliable reasoning and planning, long-running agents, multimodal and world models, embodied AI and robotics, continual learning and memory, smaller and more efficient models, AI for science, and improved safety and evaluation.

## Generation versus action

Generation means producing information or content such as text, images, audio, video or code. Action means changing the state of software or the physical world - for example by sending a message, editing a record, making a booking, purchasing something or operating equipment.

The distinction matters because the consequence of an error changes when a system can act. A mistaken sentence can often be corrected; an autonomous action may already have changed something in the world.

## Week 1 - big ideas

These are the durable ideas to retain even as current products, benchmarks and company names change.

- AI is a broad field, not a single product or technology.
- Learned AI differs from software whose behaviour is explicitly specified rule by rule.
- A model, an application and a wider AI-enabled system are different things.
- Capability depends not only on the model, but also on access, tools, permissions and the surrounding system.
- Benchmark scores provide evidence about selected tasks, not a complete measurement of intelligence.
- Performance, understanding, agency and consciousness are separate claims.
- AI is also an economic and institutional project, so incentives matter when interpreting claims.
- Forecasts become more uncertain as they accumulate assumptions about future technology, infrastructure, economics and society.
- AI literacy means being able to interrogate a new system and a new claim - not merely recognise today's tools.